

Solutions to multivariate fixed-point equations with coefficients given by nonnegative random matrices

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Let $(T_1, T_2, \dots)$ be a sequence of random $d \times d$ matrices with nonnegative coefficients. We consider a fixed-point stochastic equation of form

$$X \stackrel{d}{=} \sum_{j=1}^N T_j X_j \quad (1)$$

where $(X_j)_{j=1}^N$ are i.i.d. copies of X , independent from $(T_1, T_2, \dots)$. $N = \#\{T_j \neq \mathbf{0}\}$ and w.l.o.g. we may assume that $T_j \neq \mathbf{0}$ for all $j = 1 \dots N$. The equation (1) has Fourier version

$$\phi_X(x) := \phi(x) = \mathbb{E} \left[\prod_{j=1}^N \phi(T_j^\top x) \right], \quad (2)$$

where ϕ is a c.f. of X .

Possible applications:

- Limiting distributions in kinetic models, see f.e. [Dolera, Regazzini (2014)], [Bassetti, Ladelli, Mathes (2015)], [Mentemeier, V. (2026)];
- various models in stochastic geometry, probabilistic combinatorial optimization, analysis of recursive algorithms etc.: see f.e. [Alsmeyer, Biggins, Meiners (2012)] for a detailed survey.

Related results in multivariate case:

- Buraczewski, Guivarch, Damek, Mentemeier (2014): Existence and properties of solutions, $\text{supp}(X) \subseteq C$ for a closed convex cone $C \subset \mathbb{R}^d$, $(T_j, X_j)_j$ i.i.d. and $\text{supp}(T_1)$ leaves C invariant.
- Iksanov, Meiners (2015): Solution set, $(T_j)_j$ are scalars in $\mathbb{R}$, $\text{supp}(X) \subseteq \mathbb{R}^d$.
- Kolesko, Mentemeier (2015): Solution set, $(T_j)_j$ are nonnegative matrices, X is nonnegative;
- Mentemeier (2016): Solution set, $(T_j)_j$ are nonnegative matrices, X is nonnegative.
- Meiners, Mentemeier (2017): Solution set, $(T_j)_j$ are similarity matrices, X in $\mathbb{R}^d$.

- Our first assumption is

(A1) $N > 1$ a.s. and $\mathbb{E}[N] \in (1, \infty)$.

- Define a measure μ on $\mathbb{S}^{d-1}$ via

$$\mu(f) \equiv \int f \, d\mu := \frac{1}{\mathbb{E}N} \mathbb{E} \left[\sum_{j=1}^N f(T_j) \right].$$

- Let $(M_i)_{i=1,\dots,n}$ be i.i.d. with law μ , define

$$m(s) := \mathbb{E}[N] \left(\lim_{n \rightarrow \infty} (\mathbb{E}[\|M_n \dots M_1\|^s])^{\frac{1}{n}} \right).$$

- We say that a semigroup Υ of $\mathbb{R}_+^{d \times d}$ satisfies condition (C), if
 - 1 each $a \in \Upsilon$ doesn't have zero row or column;
 - 2 Υ contains a positive matrix;

Let Υ be a subsemigroup of $\mathbb{R}_+^{d \times d}$ generated by $\text{supp}(\mu)$. Our further assumptions are:

(A2) There exists $\alpha \in (0, 1) \cup (1, 2)$ such that $m(\alpha) = 1$ and $m'(\alpha) < 0$;

(A3) Define $\iota(a) := \inf_{x \in \mathbb{S}_+^{d-1}} \|ax\|$. Assume

$$\mathbb{E}[\|M\|^\alpha \log(1 + \|M\|)] < \infty, \quad \mathbb{E}[\|M\|^\alpha |\log(\iota(M^\top))|] < \infty$$

(A4) Υ satisfies condition (C);

(A5) Υ is strongly irreducible: $\Upsilon V \not\subseteq V$ for every proper subspace $V \subset \mathbb{R}^d$;

(A6) Υ consists of invertable matrices.

We further impose the condition on X :

(C1) $\text{supp}(X) \subseteq \mathbb{R}_+^d \cup \mathbb{R}_-^d$ and $\exists C > 0$ s.t. $\mathbb{P}(\|X\| > t) \leq Ct^{-\alpha}$ for all $t > 0$.

- Let $\mathbf{T} = \bigcup_{n \in \mathbb{N}_0} \mathbb{N}^n$ denote the infinite Ulam-Harris tree with the root $\emptyset$.
- For a node $v = v_1 \dots v_n \in \mathbf{T}$ we say, that v is in the n -th generation and write $|v| = n$.
- To each node in $v \in \mathbf{T}$ we attach an i.i.d. copy $(T_1(v), T_2(v), \dots)$ of $(T_1, T_2, \dots)$. Let $\mathbf{B} := (T_1(v), T_2(v), \dots)_{v \in \mathbf{T}}$ be a family of i.i.d. copies of the sequence $(T_1, T_2, \dots)$. We call $\mathbf{B}$ the weighted branching process rooted at $\emptyset$.
- For each node $v \in \mathbf{T}$ we define recursively

$$L(\emptyset) := \mathbf{Id}, \quad L(vj) := L(v)T_j(v)$$

where $\mathbf{Id}$ denotes a $d \times d$ identity matrix.

- Denote by $F := (F_n)_{n \in \mathbb{N}_0}$ the natural filtration of $\mathbf{B}$: for any $n \in \mathbb{N}_0$ $F_n := \sigma(L(v), |v| \leq n)$. Denote by $F_\infty := \lim_{n \rightarrow \infty} F_n$.
- For a random element E we denote by $\{[E]_v : v \in \mathbf{T}\}$ a set of i.i.d. copies of E associated with $\mathbf{T}$.

- Denote $a \cdot x := \frac{ax}{\|ax\|}$ for any $a \in \mathbb{R}_+^{d \times d}$ and $x \in \mathbb{R}^d$. Let $M \sim \mu$. For α in (A3) define *transfer operator* on $C(\mathbb{S}^{d-1})$:

$$P^\alpha f(u) := \mathbb{E}[|Mu|^\alpha f(M \cdot u)] \quad (3)$$

and denote by $(P^\alpha)'$ its dual operator acting on $M(\mathbb{S}^{d-1})$.

- Consider $(\tilde{P}^\alpha)' \nu(\cdot) := \frac{(P^\alpha)' \nu(\cdot)}{(P^\alpha)' \nu(\mathbb{S}^{d-1})}$ acting on $\mathcal{P}(\mathbb{S}^{d-1})$. Since $\mathbb{S}^{d-1}$ is compact, by Schauder-Tychonoff theorem there exist (non-unique) fixed points of $(\tilde{P}^\alpha)'$, which are then eigenmeasures of $(P^\alpha)'$.
- In [Buraczewski et.al. (2014)] it has been proved, that the restriction of $(P^\alpha)'$ to $\mathbb{S}_+^{d-1}$ has a unique eigenmeasure ν_+^α with eigenvalue equal to $m(\alpha) = 1$. By using similar techniques and the result in [Hennion (1997)], we prove

Lemma 1

Let ν^α be an eigenmeasure of $(P^\alpha)'$. Then

- its eigenvalue is equal to $m(\alpha) = 1$;
- $\text{supp}(\nu^\alpha) \subseteq \mathbb{S}_+^{d-1} \cup \mathbb{S}_-^{d-1}$;
- $\nu^\alpha = \lambda_+ \nu_+^\alpha + \lambda_- \nu_-^\alpha$. Here ν_+^α and ν_-^α are unique eigenmeasures of the restricted operator $(P^\alpha)'$ on $\mathbb{S}_+^{d-1}$ and $\mathbb{S}_-^{d-1}$, and $\lambda_+, \lambda_- > 0$ are arbitrary constants.

Theorem 1

Assume (A1)–(A6). If X is a solution to (2) satisfying (C1), then:

(i) For any $x \in \mathbb{R}^d$ the c.f. of X has form

$$\phi(x) = \mathbb{E} \left[\exp \left\{ i \langle x, Z \rangle - \int_{\mathbb{S}^{d-1}} |\langle x, y \rangle|^\alpha \left(1 - i \tan\left(\frac{\pi\alpha}{2}\right) \text{sgn}(\langle x, y \rangle) \right) \theta(dy) \right\} \right]$$

where θ and Z are F_∞ -measurable a.s. finite measure on $\mathbb{S}^{d-1}$ and r.v.

(ii) θ and Z satisfy

$$Z = \sum_{|v|=1} L(v)[Z]_v, \quad \int_{\mathbb{S}^{d-1}} f(y) \theta(dy) = \sum_{|v|=1} \int_{\mathbb{S}^{d-1}} \|L(v)y\|^\alpha f(L(v)y) [\theta]_v(dy) \quad \text{a.s.}$$

Moreover, $Z = 0$ a.s. if $\alpha \in (0, 1)$.

(iii) $\bar{\theta} := \mathbb{E}\theta$ is an invariant measure of $(P^\alpha)'$, and it holds either $\text{supp}(\bar{\theta}) \subseteq \mathbb{S}_+^{d-1}$ or $\text{supp}(\bar{\theta}) \subseteq \mathbb{S}_-^{d-1}$.

If $\alpha \in (0, 1)$, the converse is true.

- For any $x \in \mathbb{R}^d$ let $(M_n(x))_{n \in \mathbb{N}_0}$ be defined as

$$M_n(x) := \prod_{|v|=n} \phi(L(v)^\top x)$$

- By (2) it is a multiplicative martingale. It is bounded by 1 in absolute value, hence by MCT it has an a.s. and mean pointwise limit $\lim_{n \rightarrow \infty} M_n(x) = M_\infty(x)$. In view of (2) it holds $\phi(x) = \mathbb{E}M_\infty(x)$.
- Observe that

$$M_n(x) = \prod_{|v|=n} \mathbb{E}[e^{i\langle X(v), L(v)^\top x \rangle} | F_n] = \mathbb{E}\left[e^{i\langle \sum_{|v|=n} L(v)X(v), x \rangle} \middle| F_n\right],$$

hence M_n is a c.f. of $\sum_{|v|=n} L(v)[X]_v$ conditional given F_n . In view of $\sup_{|v|=n} \|L(v)\| \rightarrow 0$ a.s., we conclude that ϕ is a c.f. of a limit of row sums in a triangular-0-array. Hence M_∞ is infinitely divisible: we have $M_\infty(x) = e^{\Phi(x)}$, where Φ is given by

$$\Phi(x) = i\langle x, \mu \rangle - \frac{1}{2}x^\top \Sigma x + \int e^{i\langle x, y \rangle} - 1 - i\langle x, y \rangle \mathbb{1}_{[0,1]}(|y|) \vartheta(dy)$$

Thank you for your attention!



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